

Animal Identification Case Study Rubric

DS 4002 – Spring 2024 - Flavien Paul Moise

Due: May 06

Submission format: Upload link to github repo to canvas

Individual Assignment

General Description: Submit to canvas a link to your Github case study repository.

Why am I doing this? This is a fantastic opportunity to learn about how to use High Performance Computing, learn about state of the art CNN architectures, and manage terabytes of data. Although this may sound very advanced and daunting, don't worry - it's much simpler than it sounds. As you work on this assignment, you will gain practical skills that help solve a real use case.

- Course Learning Objective: train and test a CNN on animal recognition accuracy

What am I going to do? First, you are going to clone this Github repository, and download and unzip the data files in a folder on the HPC cluster, as instructed in the data document. Then, you are going to research the ResNet50 model architecture, and look at the code necessary to train it. After learning about that, you are going to write code yourself in a python notebook (.ipynb), which will be composed of two parts: data preprocessing and labelling, and the model itself. The data preprocessing needs to associate every image with a label, and then re-size the image to be acceptable to the model's input layer. The model should be done in PyTorch, print the results at the end to the terminal, and take advantage of GPUs if present. Once your code is done, run it for 10 minutes normally (without GPUs) to ensure that it doesn't have errors in it. Then, export the code as a Python file (.py), and create a SLURM job for it. Finally, you are going to run the SLURM job with a GPU, wait a day or so while it runs, and then look at the results.

Your final deliverables should include

- Well documented code you wrote to process data, and train and test the model
- The model's output and accuracy
- The SLURM job script you ran
- A Github repository containing your work.

All of this will be submitted electronically via a link to a github repository.

Tips for success:

- Start early. Downloading the data, unzipping it, and running the model can take many hours.
- Don't train a ResNet50 algorithm from scratch. Use a pretrained model as the base weights to take advantage of transfer learning.
- Familiarize yourself with Python and PyTorch beforehand. You should know the general structure of torchvision models and syntax.
- Talk to the instructors for help. They can help you with HPC issues, SLURM jobs, and any other issues that arise.

- Communicate with your team to ensure you are all on the same page. This project has components that take a while to run: you might not be able to re-run a program if you realize you made a communication mistake early on.

How will I know I have Succeeded? You will meet expectations of this case study when you follow the criteria in the rubric below.

Spec Category	Spec Details
Formatting	• One Github repository, submitted via link on Canvas. • Folder Structure ▪ Data/ ▪ Should contain a data dictionary .pdf ▪ Scripts/ ▪ .ipynb script ▪ .py version of the .ipynb script ▪ SLURM job script ▪ Output/ ▪ Should contain the result of the SLURM job ▪ README.md ▪ LICENSE.md
Data Dictionary	• <u>Goal</u>: This pdf should explain what data is used, and how to download it to an HPC cluster. - o The data is too large to be hosted on Github. • Include step by step instructions on what you did download the data. • Explain the data folder structure and what is in each type of file.
Scripts	• <u>Goal</u>: Well documented code that contains the data processing steps and the training steps. There should also be a SLURM job script. • The code should be developed in an .ipynb file, but run in a .py file. The code in each file should be well documented and identical. • The SLURM job should be a bash .sh script that runs the .py file above.
README.md	• <u>Goal</u>: Overview of the project • Should explain the structure of the Github repository you're submitting. • Explain commands should be run to get the results you did. • Should list any references you used at the bottom in IEEE documentation style.
LICENSE.md	• <u>Goal</u>: License for the use of your project • MIT License is fine, or any other license that you think is necessary.

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